PROJECT ARTEMIS -X

SUSTAINABLE LUNAR SOUTH POLE ARCHITECTURE | LUNAR BASE DESIGN CHALLENGE

• OVERVIEW

ACTIVE

Verified Identity

Debjit Dutta

PRIMARY ROLE

AI Project Manager & Concept Lead (Simulated 'Chief Systems Engineer' for the challenge)

SECURITY ACCESS
SYSTEM STATUS

NOMINAL // CONNECTED

PROJECT ARTEMIS -X

SUSTAINABLE LUNAR SOUTH POLE ARCHITECTURE | LUNAR BASE DESIGN CHALLENGE

• PROJECT OBJECTIVE

MANAGEMENT

Orchestrating AI to Build the Future

To demonstrate technical project management, rapid prototyping, and AI orchestration by acting as a director for AI coding assistants, rather than hand-coding a 3D environment from scratch or engineering real-world aerospace hardware. Here is how management shaped this design:

Prompt Engineering

Translated strict aerospace design constraints into programmatic prompts to guide AI code generation.

Asset Integration

Managed the integration of multiple AI-generated geometries (Starships, Solar Towers, Kevlar Domes) into a cohesive WebGL scene.

Quality Assurance

Acted as the technical reviewer, debugging and refining the AI-generated JavaScript to ensure a performant browser experience.

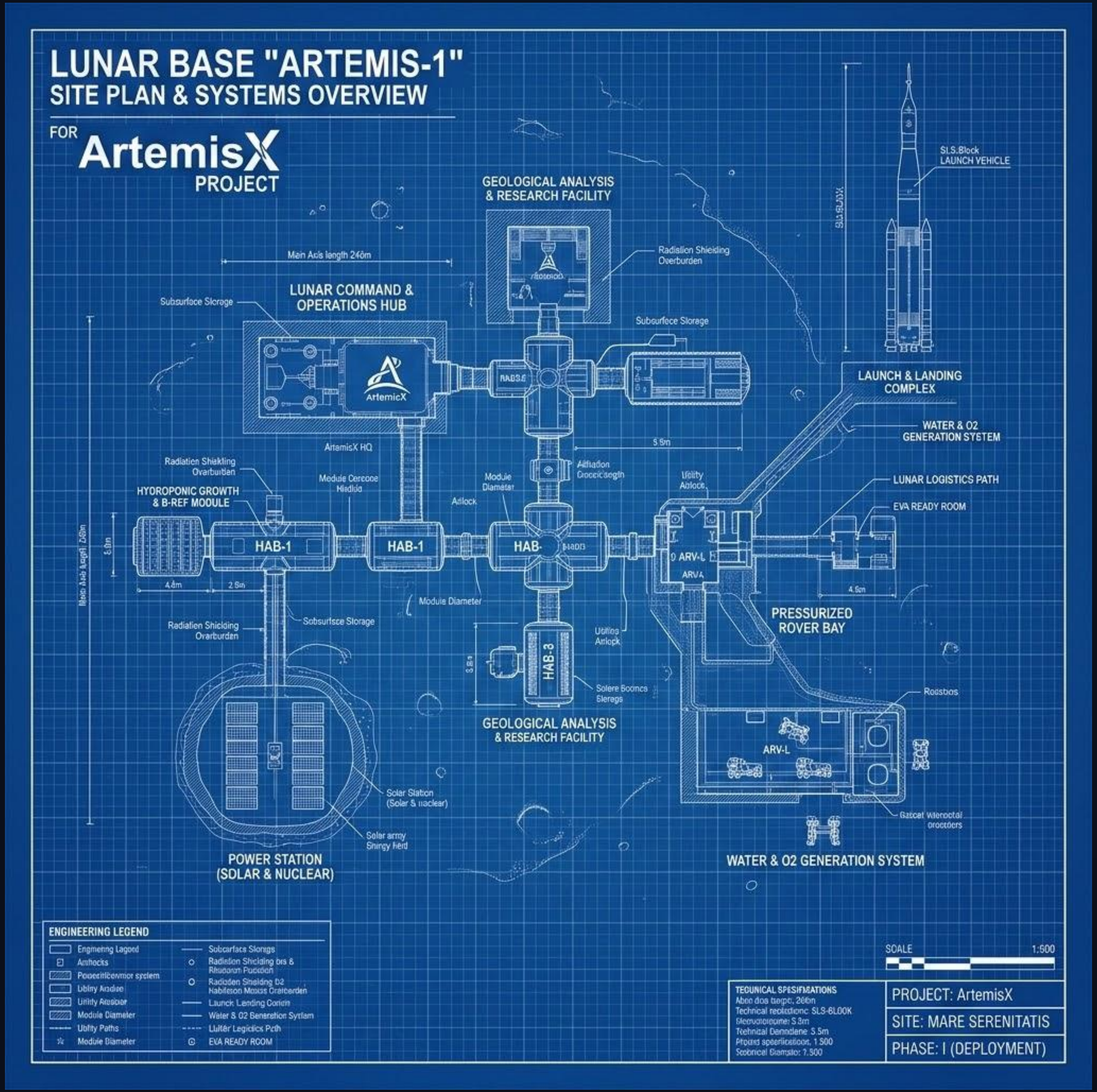
Mission Overview

Strategic Location

Situated at the South Pole's Shackleton Crater, our architecture exploits near-constant solar illumination and proximity to permanently shadowed regions (PSRs) rich in water ice.

Zero-Waste Paradigm

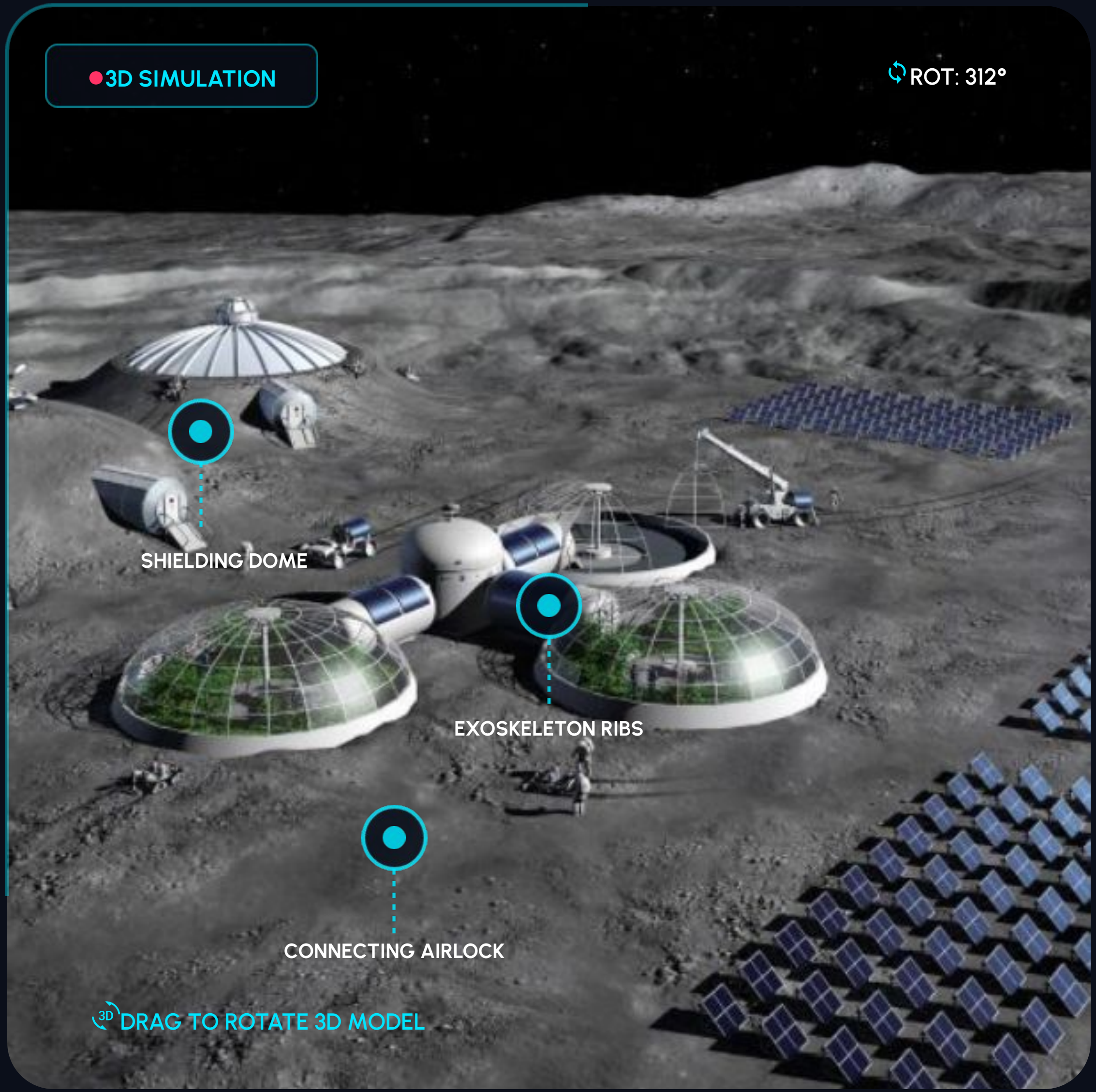
Operating on a strict closed-loop ecosystem, the base utilizes advanced In-Situ Resource Utilization (ISRU) to achieve unprecedented Earth-independence and sustainability.



Structural Architecture

Inflatable Kevlar Habitats

- **Autonomous Shielding:** Rovers sinter and pile 2-3 meters of lunar regolith over domes to block Galactic Cosmic Rays and micrometeoroids.
- **Exoskeleton Ribs:** 3D-printed external load-bearing ribs distribute the massive weight of the regolith shielding.
- **Zenith Nodes:** Rigid top-mounted caps house optical sensor suites and emergency vertical docking ports.
- **Reinforced Airlocks:** Interconnecting pressure corridors feature ribbed reinforcement to prevent fatigue and depressurization.



INTERACTIVE 3D SIMULATION

Experience Project Artemis-X in Real-Time

Step inside the digital twin of our proposed lunar South Pole base. This high-fidelity, interactive 3D blueprint allows you to fully inspect the structural integration, solar arrays, and sub-surface architectures.

Key Interactive Features

■ **Orbital Rotation:** Drag to spin the entire base model and examine the structural distribution.

■ **Cross-Section Views:** Cut away the 3-meter sintered regolith shields to view interior Kevlar domes.

■ **Asset Inspection:** Click on individual components like the Zenith Sensor Nodes or vertical solar towers for live telemetry telemetry specifications.

SCAN TO ROTATE MODEL

https://debjit-prime.github.io/ARTEMIS-X/

Compatible with iOS and Android devices.
Instantly launch the WebGL interactive experience.

STRUCTURAL INTEGRATIONS

Autonomous Shielding

Prior to crew arrival, rovers sinter and pile 2-3 meters of lunar regolith over the inflatable Kevlar domes, providing vital protection against Galactic Cosmic Rays (GCRs).

Exoskeleton Ribs

The habitat domes feature 3D-printed external structural ribs that evenly distribute the massive, crushing weight of the regolith shielding across the base foundation.

Zenith Sensor Nodes

Each dome is capped with a rigid zenith node housing optical sensors, emergency ventilation valves, and vertical docking ports for rovers and landing modules.

ENERGY & HYDRATION SYSTEMS

1. Power Generation

Vertical Solar Towers: Due to the low polar sun angle, standard flat panels fail. We deploy towering bi-facial arrays that track the sun across the horizon.

Storage & Failsafe: Regenerative Fuel Cells (RFCs) recombine hydrogen and oxygen at night. A 10kW buried Kilopower fission reactor provides a constant emergency baseline load.

2. Water Recovery

Ice Sublimation: Autonomous thermal-mining rovers extract water ice from the crater floor, sublimating it into vapor and leaving toxic lunar dust behind.

Closed-Loop ECLSS: Water is too heavy to import. We recover 98% of all habitat moisture—including sweat and respiration—via catalytic oxidation and distillation.

ISRU & Biosystems Integration

3. Oxygen Prod.

Lunar regolith is ~45% oxygen by weight. **Molten Regolith Electrolysis (MRE)** heats soil to 1600°C. An electrical current separates pure breathable oxygen from the magma, leaving useful metal alloys behind.

4. Food (Aeroponics)

Food is cultivated in vertical aeroponic greenhouses using 95% less water than soil farming. Narrow-band red and blue LEDs supply exact Photosynthetically Active Radiation (PAR) to minimize power draw.

5. Cyanobacteria

Photobioreactors cultivate *Spirulina* (microalgae). It rapidly consumes crew CO_2 output and produces a highly dense, easily harvested source of protein and essential vitamins in low-gravity.

Advanced Infrastructure

6. Waste Management

The Sabatier Reaction recovers atmospheric gases, combining Carbon Dioxide and Hydrogen to yield Methane (for lander propellant) and Water:

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

Solid biological waste is aerobically processed into aeroponic fertilizer, closing the loop.

7. Communication Relay

Primary Uplink: Direct-to-Earth (DTE) Optical (Laser) Communications provide gigabit data rates for real-time telemetry and telemedicine.

Relay Architecture: To prevent blackout periods caused by lunar rotation, a dedicated Lunar Relay Satellite in a Near-Rectilinear Halo Orbit (NRHO) establishes 24/7 telecommunications redundancy.

PAYLOAD MASS ALLOCATION

⚡ Telemetry: Active

Target: 85% ISRU

By relying on **In-Situ Resource Utilization (ISRU) for 85%** of the total base mass, Project Artemis-X drastically reduces Earth-launch payload costs and achieves unprecedented mission sustainability.

Artemis-X Deployment Timeline

PHASE 01 · ACTIVE

Preparation

Autonomous rovers deploy to Shackleton Crater to sinter regolith foundations and establish initial vertical solar array grids.

✓ ROVERS DEPLOYED

PHASE 02 · QUEUED

Construction

Inflatable Kevlar habitats land and pressurize. Rovers bury the modules in 3 meters of regolith shielding.

⚡ READY FOR LANDING

PHASE 03 · PENDING

Integration

ECLSS, Aeroponics, and Molten Regolith Electrolysis systems are brought online and tested autonomously.

○ ECLSS STAGED

PHASE 04 · PENDING

Operations

First crewed landing. Astronauts enter a fully functional, pressurized, and thermally stable biosphere.

○ CREW ASSIGNED

Conclusion

Project Artemis-X proves humanity can establish a permanent foothold through sustainable, closed-loop environmental design.

National Space Day: Lunar Base Design Challenge

Mission Ready.

Establishing a permanent, self-sufficient foothold on the lunar surface.

Designed for the National Space Day Lunar Base Challenge